

Topic 3 of Personnel Economics: Pay for performance in practice

Exercises

Exercise 3.1

A manager of a local production unit is thinking about prohibiting workers to snapchat during work time. Snapchatting yields a utility gain expressed in euros to the workers of V per worker.

Snapchatting yields a production value loss expressed in euros to the manager of X per worker.

Assume initially that it is costless for the manager to monitor whether workers snapchat, and that the manager can impose a non-monetary penalty on workers who do so, which is costless for the manager, but yields a utility loss of F to the worker. We assume that $F > V$.

- a) Is it socially efficient to allow workers to snapchat?
- b) Suppose the workers earn a rent which is large enough so that they never want to quit this job. What will the manager decide on allowing snapchatting by the workers?
- c) Suppose wages are set by the manager so that workers are always indifferent between working for the manager and working somewhere else. What will the manager decide on allowing snapchatting by the workers?
- d) Now suppose it is not so easy to monitor workers. The manager needs to invest monitoring cost C per worker in order to detect whether workers have snapchatted. Answer questions a), b), and c).
- e) How do the results change when $F < V$?

Exercise 3.2

Consider a worker who works for at most two periods (say, when young and when old) at a particular firm. Each period, the worker can choose between exerting high effort and exerting low effort. The worker's cost of exerting high effort are c_y when young and c_o when old. The worker's cost of exerting low effort are zero. The effort of the worker is difficult to observe by the firm. With probability π , the firm observes the worker's effort at the end of a period. With the remaining probability, the firm does not observe the worker's effort. The firm can only fire the worker when it observed that the worker has shirked. In that case, the firm is also allowed to withhold the worker's wage for the period in which the worker was caught shirking. When exerting high effort, the worker's productivity is q_y when young and q_o when old. When exerting low effort, the worker's productivity is zero. The worker's value of alternative job opportunities is v_y when young and v_o when old. For simplicity, the discount rate is assumed to be zero. We also assume that the worker cannot work elsewhere when young and after that work at this firm when old.

- a) What is the minimum level of the wage for which an old worker will choose to work at this firm and will exert high effort? Interpret your result.

Suppose the wage in the second period is such that the worker will choose to work at this firm and will exert high effort.

b) What is the minimum level of the wage for which a young worker will choose to work at this firm and will exert high effort? Interpret your result.

Suppose $\pi=0.2$, $c_O = c_Y = 100$, and $v_O = v_Y = 200$.

c) What is the minimum level of wages in period 1 and 2 so that the worker exerts high effort in both periods and is willing to work at this firm in both periods?

A new technology has been invented which makes it easy for the firm to monitor the worker. When the firm applies the technology, it observes worker's effort each period with probability one.

d) What is the minimum level of wages in period 1 and 2 when the firm applies the new technology?

e) How much is the firm willing to pay for the new technology?

Chapter 7, discussion questions 1, 2, 3, 4.

Exercise 3.3 (partly based on an Exam question from last year)

In the book a study is discussed about working from home. The study is a field experiment (also known as a randomized controlled trial or RCT). The researchers report that the average productivity of workers who started working from home went down. Nevertheless, they conclude that working from home increased productivity of workers.

a) Explain in one sentence how both these things can be true at the same time. (If you write more than one sentence, we will only consider your first sentence.)

b) A well-known problem with empirical studies using observational data is reversed causality. Explain why reversed causality is not a problem when you run a field experiment.

Answers

3.1

- a) Snapchatting increases social welfare by $V - X$. Hence, if $V > X$, it is socially efficient to allow it.
- b) The manager will not allow it. This way, her payoff increases by X . The workers utility decreases by V , but they won't quit. The penalty F will induce the workers to comply because the detection probability is 1 and $F > V$.
- c) The manager will allow it if $V > X$. Not allowing it would raise production value by X , but increase wages by V .
- d) Allowing snapchatting now also saves on the monitoring costs and so is socially efficient if $V - X > -C$. If workers earn a rent, the manager will allow it if $-X > -C$, that is when $C > X$, i.e. if the monitoring cost are too high. If workers do not earn a rent, the manager will allow it if $V - X > -C$.
- e) If the maximum penalty that can be imposed on a worker is lower than the value of snapchatting to the worker, the worker cannot be kept from snapchatting. Hence, the manager will always allow it, even if it is socially inefficient.

3.2

- a) To induce the worker to exert high effort, it must hold that $\pi \cdot 0 + (1 - \pi)w_O < w_O - c_O$, implying $w_O > c_O / \pi$. To make it attractive for the worker to work at this firm, the wage must be greater than or equal to $v_O + c_O$. Note that when $v_O = 0$, then if the first condition is satisfied, the second is as well.
- b) To induce the worker to exert high effort, it must hold that $\pi(0 + v_O) + (1 - \pi)(w_Y + w_O - c_O) < w_Y - c_Y + w_O - c_O$, implying $w_Y > c_Y / \pi - (w_O - c_O - v_O)$. To make it attractive for the worker to work at this firm, the wage must be such that $w_Y - c_Y + w_O - c_O > v_Y + v_O$, or: $w_Y > c_Y + v_Y - (w_O - c_O - v_O)$. Note that it depends on the values of π and v_Y whether one or the other restriction is binding.
- c) Use the results under a) and b). In period 2, the wage must be equal to 500. In period 1 to 300. Interpretation: In period 1 it is easier to induce high effort as more is at stake than in period 2.
- d) It is $200 + 100 = 300$ in both periods.
- e) The cost saving is $500 + 300 - 300 - 300 = 200$. The firm's maximum willingness to pay is thus 200.

Answers to Chapter 7, discussion questions 1, 2, 3, 4.

Reading chapter 7 carefully will enable you to provide answers to these questions. Discuss in class.

3.3

a) If the productivity in the control group went down even more, the conclusion is correct that working from home increased productivity.

b) Reversed causality means that the dependent variable in a regression x may not only affect the dependent variable y , but y may also affect x . As a result, looking at the correlation between x and y does not tell you anything (or at least not a lot) about the effect of x on y . For instance, in the context of personnel economics, the presence of bonus pay may cause high performance, but it could as well be that high performance results in the presence of bonus pay. In a field experiment the problem of reversed causality never shows up, because the variation in x (e.g. whether or not a bonus can be earned) stems from random assignment.